COMP-8677 Project

The objective of the project is to build your self-learning and team work capability. It also gives an opportunity to conduct research and build your research capability. You need to form a team of at most **5 members** and the deadline of your report is **Aug 6**. *Register your team members on bright space. Email me your project choice (if it is a paper, send me the paper), your team ID on brightspace and the team members.* **If you do not register your name in your team on brightspace, your project mark will be decided as zero, regardless of your real contribution to the project.** The project should be done as a team work. Your team members now should not be changed without the instructor's permission. You have three options for the project. You need to tell me your option by **June 20**.

1. **Option 1 (research)**: choose a paper **published after 2010** from one of the venues: NDSS, PoPETs, IEEE ICBC, IEEE TIFS, IEEE TDSC, AsiaCCS, CCS, IEEE S&P.
   IEEE TIFS: https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=10206
   IEEE TDSC: https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=8858
   IEEE S & P: https://dblp.org/db/conf/sp/index.html
   AsiaCCS: https://dblp.org/db/conf/asiaccs/index.html
   CCS: https://dblp.org/db/conf/ccs/index.html
   NDSS: https://dblp.org/db/conf/ndss/index.html
   IEEE ICBC: https://dblp.org/db/conf/icbc2/index.html
   PoPETs: https://dblp.org/db/journals/popets/index.html

You can also choose one paper from other conferences but up to the instructor's approval. After you select the paper of your focus, send the file to me. Note one paper can only be approved to at most one group. For conferences, please do not choose the paper from the *affiliated* workshops.

Project Requirement: you need to focus on your selected paper and show your understanding on it. Present in **details** how the main contribution works. For example, describe, explain and analyze the algorithm and protocol in the paper. Examples or implementation will be useful for this purpose. DO NOT give me the overall description of the contribution without any details. I will give you very low mark if you do this way. After reading your report, I should be able to describe the main idea of the paper. Your understanding includes clarifying the unclear or difficult details or giving some opinions on the claim in the paper (for example, the view in paper is correct/wrong, good/bad, interesting/trivial and give your reason) and (if possible) some programming experiment are encouraged. Any meaningful programming experiment will be encouraged. Anything related to the paper will count. But do not copy programs from other sources. For your report, you should NOT copy any sentence of the paper. You need to understand the content first and then write it in your own language.

2. **Option 2** (*programming project*). You can propose a programming project that you want to explore. The topic can be any but it has to be security related. You need my approval before you start. Any attempt for this type of project will be encouraged. Requirements: source code and report on your project documentation (introduction to your project, design and how they are implemented and what is the performance).
3. **Option 3** (*seedlab learning*). In seedlab website, there are a lot of security experiments. We have only learned only a few of them. In this option, you can choose one new lab to learn. Requirements: write a report and demonstrate to me how it works.
4. **General requirements for all options:**
   a) Project has 16 points in total (8 points for **report** and 8 points for **presentation**). Each team will have 15 minutes to discuss the details of your project in your week 12 class. So the presentation is essentially a research discussion. You need to write PPT to make the discussion efficient. Exceptional projects will be given up to 3 bonus points.
   b) Your project file should be at least 10 pages and at most 25 pages (page style: one column, 1.0 line spaced with font size 14. This style is the same as *this guideline* you are reading).
   c) For your submission, all files should be submitted to brightspace. However, it you have a zip file (e.g., your source code), you probably are unable to submit there. In this case, email your zip file directly to me.
   d) **Warning**: DO NOT generate the report using AI tools. If I suspect you have done this, I will ask you to talk with me how you wrote this. A confirmed report from AI will result in a zero grade in this course for all of your **group** members.